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Installing And Configuring Windows Server

Windows Server Selection and Installation

1.1 Selecting appropriate windows server edition:

When planning a server deployment, we should choose the operating system edition based on multiple factors, including the following:

- The roles we intend the servers to perform
- The virtualization strategy we intent to implement
- The licensing strategy we plan to use

Compared to the numerous editions available in Windows Server 2016, Microsoft has significantly simplified the selection process for Windows Server 2022 by streamlining the available options. While still requiring a 64-bit processor architecture, unlike older versions, Windows Server 2022 offers three core editions:

Windows Server 2022 Standard Edition:

Active Directory Domain Services (AD DS): Creates a central directory for user accounts, computers, and resources, perfect for managing a small office network.

DHCP Server: Automatically assigns IP addresses to devices on your network, saving you manual configuration hassle.

File Server: Allows centralized storage and sharing of files among users in your office.

Group Policy Management: Enforces consistent configurations and security settings across all your office computers.

Windows Server 2022 Datacenter Edition:

All features of Standard Edition: Plus, additional features for larger deployments, such as unlimited Hyper-V VMs and support for more physical cores. (Not needed for small environment)

Shielded VMs: Creates secure virtual machines for sensitive workloads. (Useful for very specific security needs, not typical for small offices)

Storage Replica: Replicates storage data across servers for disaster recovery. (More relevant for larger environments with critical data)

Windows Server 2022 (Standard and Datacenter):

Similar roles to Windows Server 2022: However, nearing end of mainstream support (December 2023), increasing potential security risks. (Not recommended due to approaching end of support)

Windows Server, version 22H2 (Standard and Datacenter):

Focus on containers: Primarily designed for deploying and managing containerized applications, less emphasis on traditional network services like AD DS. (Not ideal for your focus on user management and file sharing)

Beside this all editions there are a lot of additional requirements to be studied and considered about:

Hardware Specifications:

- Processor: Confirm the number of physical cores in your server hardware. Windows Server 2022 Standard edition supports a maximum of two physical CPUs with unlimited virtual cores.
- RAM: Identify the total amount of RAM available. Minimum requirements vary depending on the features you plan to use.
- Storage: Consider the storage capacity needed for user data, applications, and virtual machines (VMs).

Office Requirements:

- Number of users: How many users will be accessing the network resources? This helps determine the licensing requirements.
- Active Directory (AD): Do you need centralized user and group management, authentication, and authorization?
- File Sharing: Will users collaborate on documents stored on a central server? You might need additional storage and file server configuration.
- Group Policy (GPO): Do you want to enforce consistent settings like security policies and software installation across all office computers?
- Virtualization: Do you plan on running virtual machines (VMs) on this server? This would require the Datacenter edition or specific licensing for the Standard edition.

For setting up a comprehensive network infrastructure for a small office environment using Windows Server . We need to select one among the three windows server editions. For small office environment **Windows Server 2022 Standard Edition** is suitable here's why:

- It covers all the features and requirements: This edition provides all the essential roles for performing the tasks organization outlined, including Active Directory, DHCP Server, File Server, Group Policy Management, and more.
- Cost-effective for small businesses: Compared to the Datacenter edition, Standard is more cost-effective for smaller deployments, making it ideal for organizations with limited budgets and resources.
- Scalability for future growth: While Standard edition has limitations on core count and virtual machines compared to Datacenter, it can still accommodate up to 2 physical CPUs with unlimited virtual cores and 2 Hyper-V VMs. This provides room for future growth in your small office environment.

Minimum hardware requirements of windows server 2022 standard edition:

Processor

- 1.4 GHz 64-bit processor
- Compatible with x64 instruction set
- Supports NX and DEP
- Supports CMPXCHG16b, LAHF/SAHF, and PrefetchW
- Supports Second Level Address Translation (EPT or NPT)

RAM

- 512 MB (2 GB for Server with Desktop Experience installation option)
- ECC (Error Correcting Code) type or similar technology, for physical host deployments

Storage controller and Disk Space

- 32GB

Network adapter requirements

- An ethernet adapter capable of at least 1 gigabit per second throughput
- Compliant with the PCI Express architecture specification.

Other requirements

- UEFI 2.3.1c-based system and firmware that supports secure boot
- Graphics device

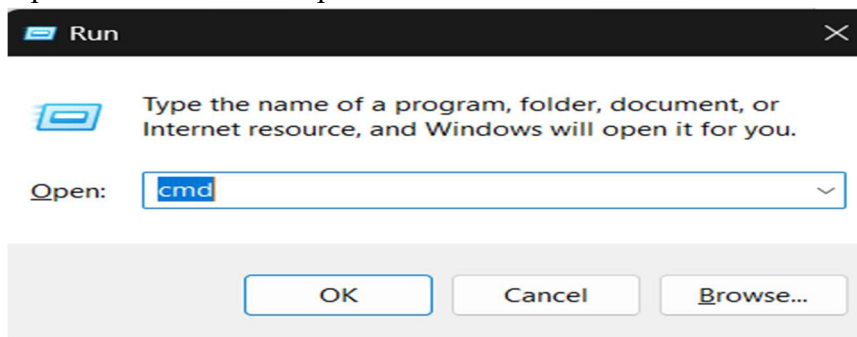
1.2 Performing the installation:

To perform a clean installation of Windows Server 2022 standard edition , use the following procedure.

1. Prepare the computer for the Windows Server 2022 installation by making sure that all its external peripheral devices are connected and powered on.
2. Turn on the computer and insert the Windows Server 2022 standard edition installation disk into the DVD drive or use Bootable USB drive for setup. The detailed steps to make storage device bootable:

To create a bootable USB flash drive:

- Insert a USB flash drive into a running computer.
- Open a Command Prompt window as an administrator.



- Type **diskpart**, and then click **ENTER**.

```
Microsoft Windows [Version 10.0.22631.3235]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NITRO>diskpart
```

- In the new command line window that opens, to determine the USB flash drive number or drive letter, at the command prompt, type **list disk**, and then click **ENTER**. The **list disk** command displays all the disks on the computer. Note the drive number or drive letter of the USB flash drive.

```
DISKPART> list disk

Disk ###  Status              Size               Free              Dyn  Gpt
-----  -
Disk 0    Online              476 GB             2048 KB           *
Disk 1    Online              7800 MB             0 B
```

- At the command prompt, type **select disk <X>**, where X is the drive number or drive letter of the USB flash drive, and then click **ENTER**

```
DISKPART> select disk 1

Disk 1 is now the selected disk.
```

- Type **clean**, and then click **ENTER**. This command deletes all data from the USB flash drive.

```
DISKPART> clean

DiskPart succeeded in cleaning the disk.
```

- To create a new primary partition on the USB flash drive, type **create partition primary**, and then click **ENTER**.

```
DISKPART> create partition primary

DiskPart succeeded in creating the specified partition.
```

- To select the partition that you just created, type **select partition 1**, and then click **ENTER**.

```
DISKPART> select partition 1

Partition 1 is now the selected partition.
```

- To format the partition, type **format fs=ntfs quick**, and then click **ENTER**. (Note: If your server platform supports Unified Extensible Firmware Interface (UEFI), you should format the USB flash drive as FAT32 rather than as NTFS. To format the partition as FAT32, type **format fs=fat32 quick**, and then click **ENTER**)

```
DISKPART> format fs=ntfs quick

    100 percent completed

DiskPart successfully formatted the volume.
```

- Type **active**, and then click **ENTER**.

```
DISKPART> active

DiskPart marked the current partition as active.
```

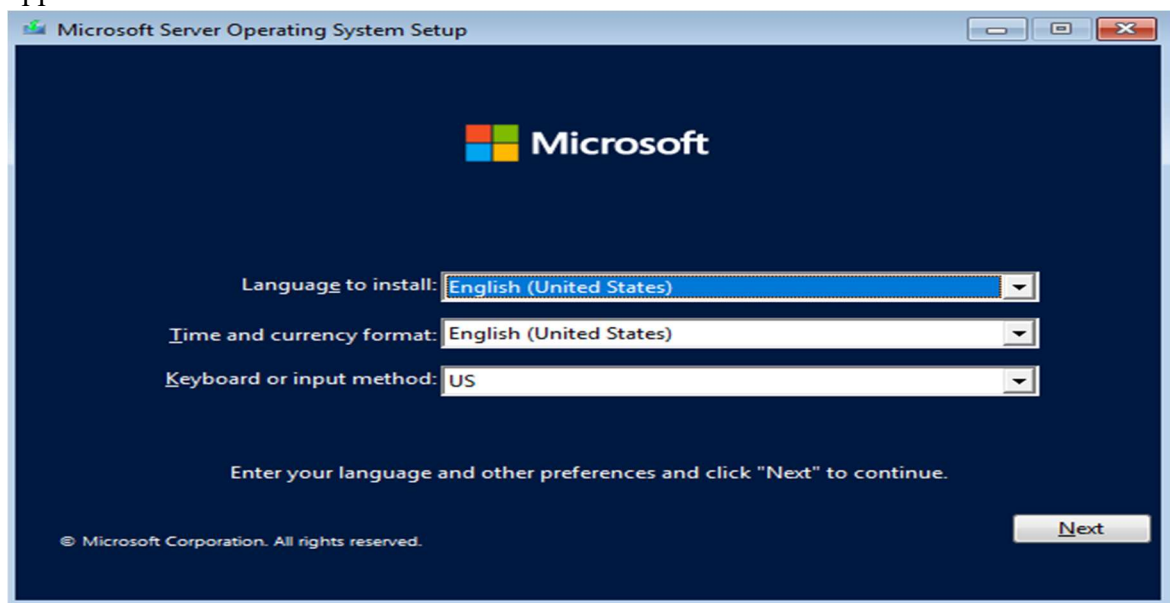
- Type **exit**, and then click **ENTER**.

```
DISKPART> exit
```

- When you finish preparing your custom image, save it to the root of the USB flash drive.

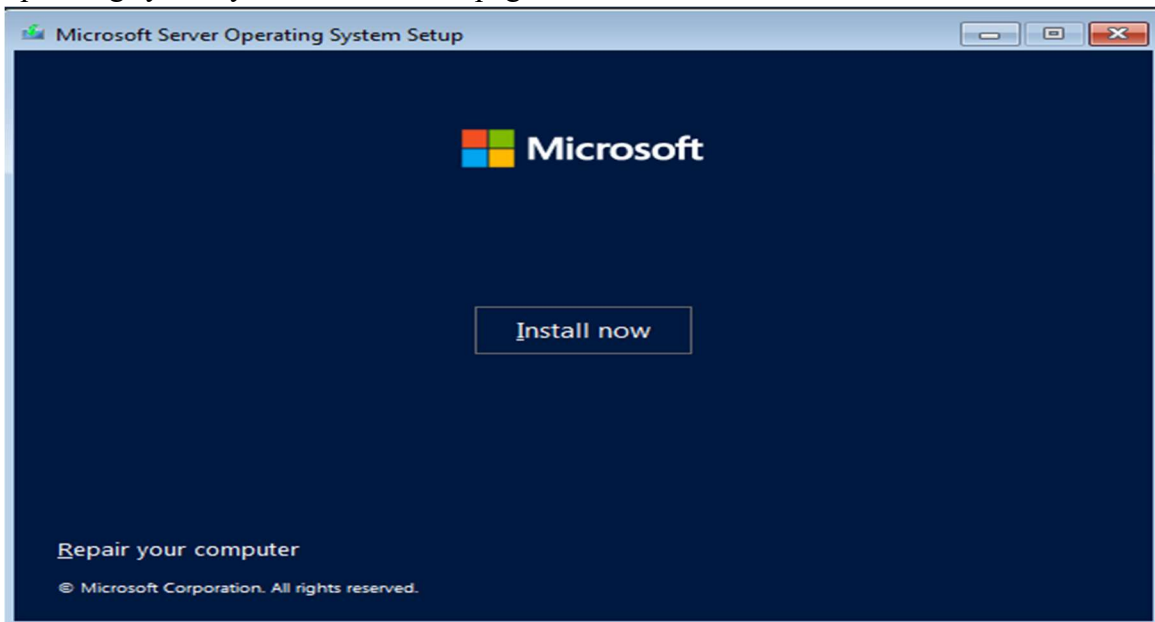
boot	12/7/2021 6:12 PM	File folder	
efi	12/7/2021 6:12 PM	File folder	
sources	12/7/2021 6:12 PM	File folder	
support	12/7/2021 6:12 PM	File folder	
autorun	12/7/2021 6:10 PM	Setup Information	1 KB
bootmgr	12/7/2021 6:10 PM	File	426 KB
bootmgr.efi	12/7/2021 6:10 PM	EFI File	1,933 KB
setup	12/7/2021 6:10 PM	Application	93 KB

3. Now the pen drive is fully bootable and consist of windows server 2022 standard edition files.
4. **Reboot** the system or PC.
5. Open the **Boot Menu**
6. Change the boot settings prioritizing the boot from External disk drive. Then save and exit.
7. Press any key to boot from the DVD (if necessary). A progress indicator screen appears as Windows is loading files.
8. The computer loads the Windows graphical interface and the Windows Setup page appears.

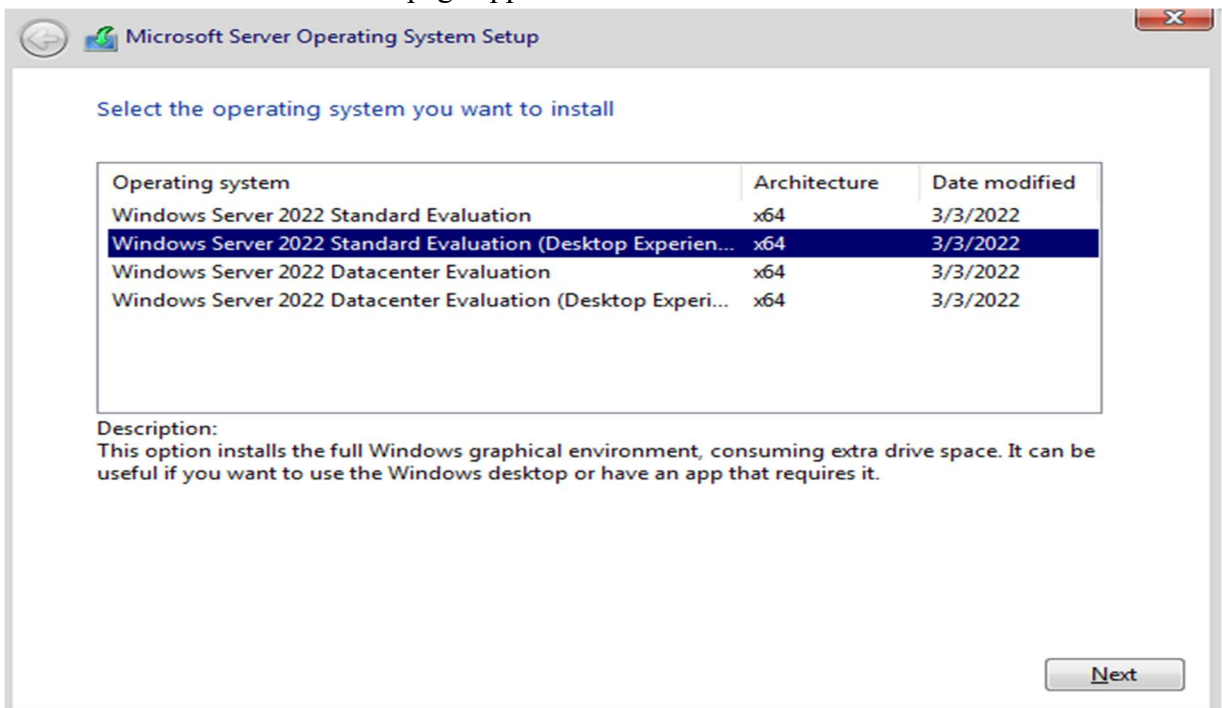


9. By using the drop-down lists provided, select the appropriate language to install, time and currency format, and keyboard or input method, and then click Next. Another Windows Setup page appears.
10. Click Install Now. The Windows Setup Wizard appears, displaying the Select the

operating system you want to install page.

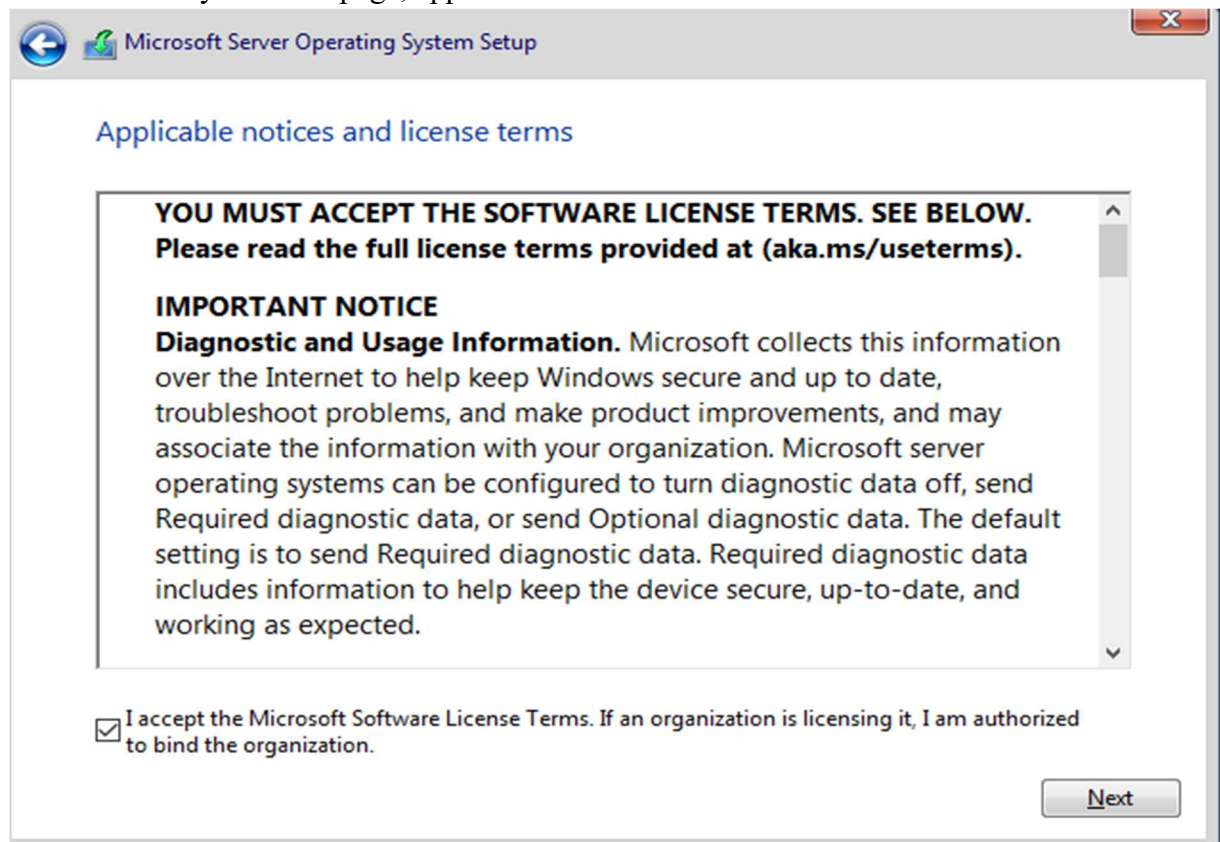


5. Select the operating system edition and installation option you want to install and click Next. The License Terms page appears.



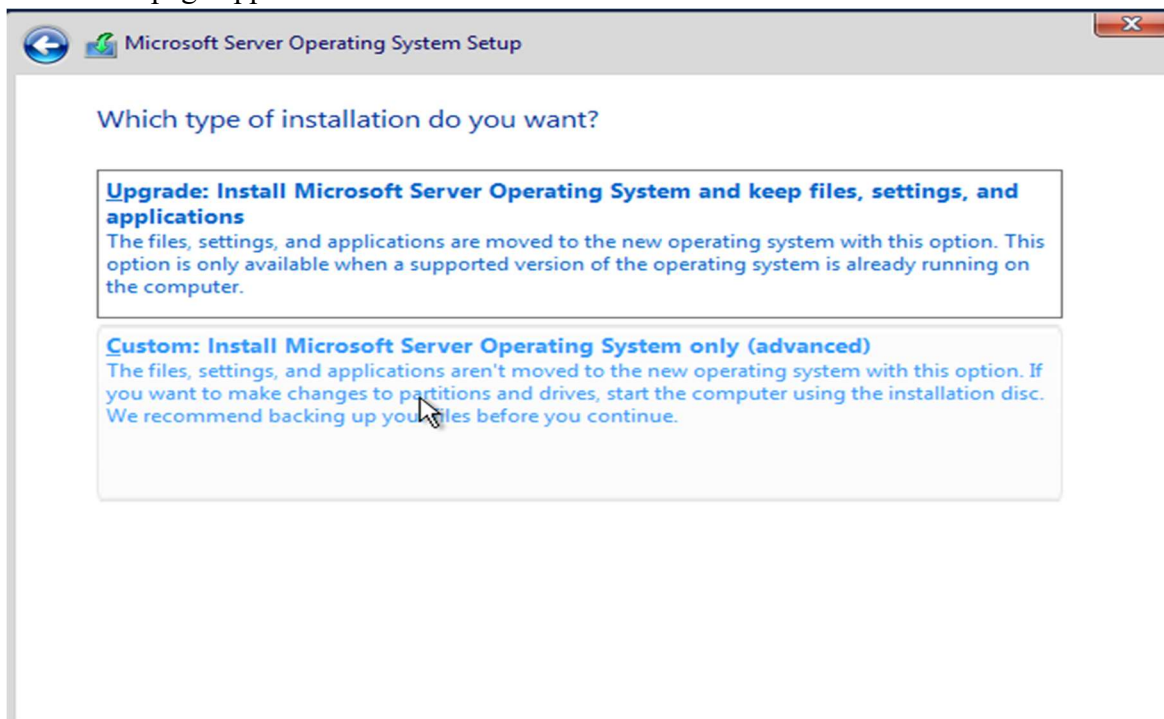
6. Select the I accept the license terms check box and click Next. The Which type of

installation do you want? page, appears.

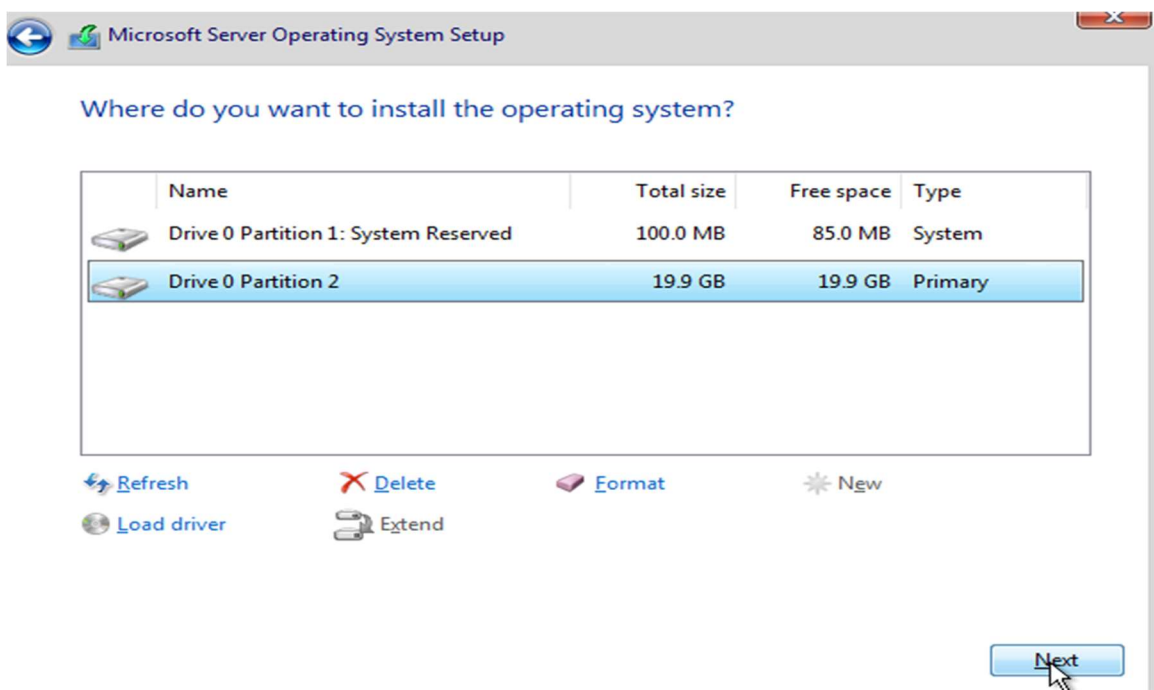


7. Because you are performing a clean installation and not an upgrade, click the Custom: Install Windows Only (advanced) option. The Where do you want to install

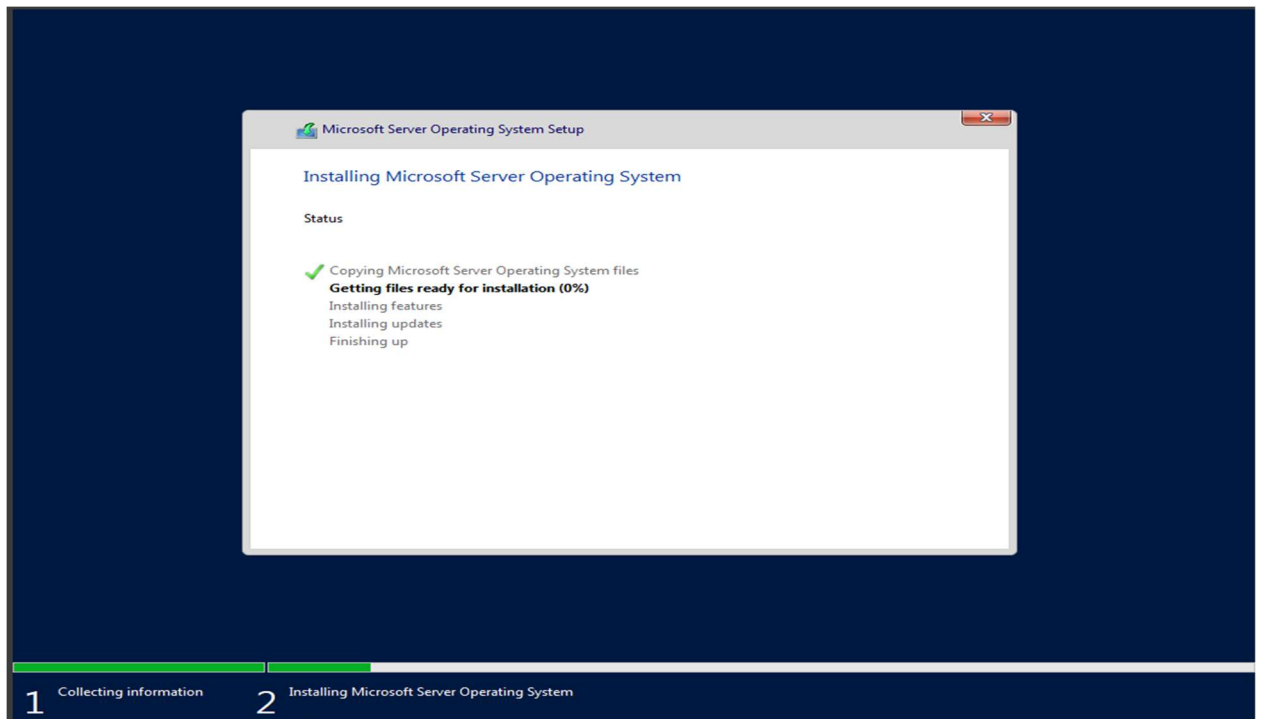
Windows? page appears.



8. From the list provided, select the partition on which you want to install Windows Server 2022, or select an area of unallocated disk space where the Setup program can create a new partition. Then select the disk and click Next.



11. The Installing Windows page appears



12. After several minutes, during which the Setup program installs Windows Server 2022, the computer restarts and the Settings page appears.

In the Password and Reenter Password text boxes, type the password to be associated with the Administrator account and press Enter. The system analyzes the installation.

13. The Windows sign-on screen appears.

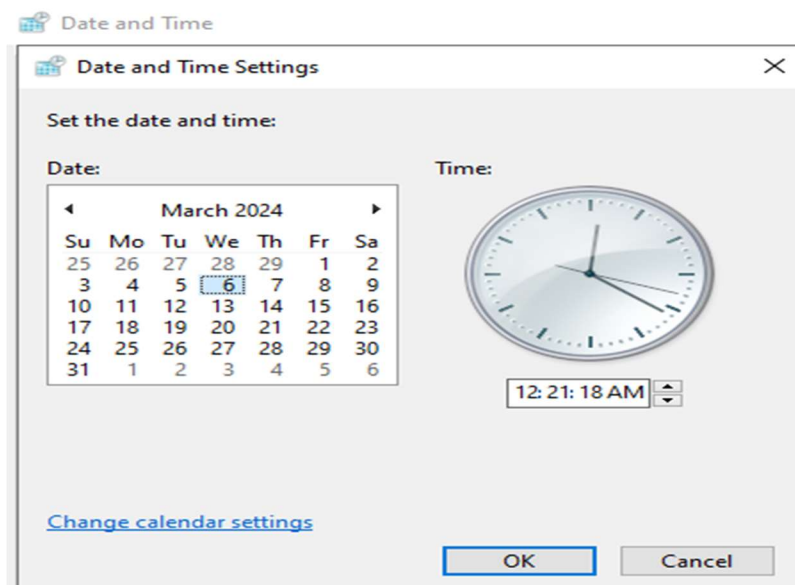


Basic Windows Configuration

2.1 Configure time zone:

Set the correct time zone for your location:

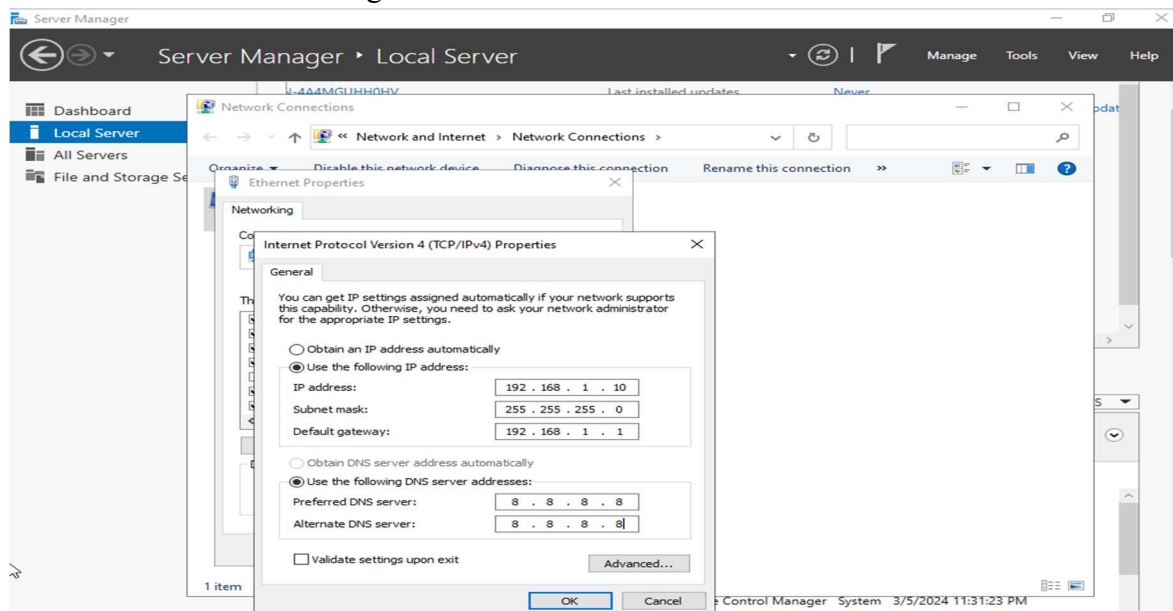
- Open Control Panel > Date and Time.
- Go to the 'Date and Time' tab and select your time zone.
- Click 'OK' to apply the changes.



2.2 Network Configuration:

Set a static IP address for the server:

- Open Control Panel > Network and Sharing Center.
- Click on 'Change adapter settings.'
- Right-click on the network adapter and select 'Properties.'
- Select 'Internet Protocol Version 4 (TCP/IPv4)' and click 'Properties.'
- Choose 'Use the following IP address' and enter the IP address, subnet mask, default gateway, and preferred DNS server.
- Click 'OK' to save the settings.



Configuration of IP Assignment using DHCP

3.1 Dynamic Host Configuration Protocol(DHCP):

The Dynamic Host Configuration Protocol (DHCP) service automatically configures the Internet Protocol (IP) address and other TCP/IP settings on network computers by assigning addresses from a pool (called a scope) and reclaiming them when they are no longer in use.

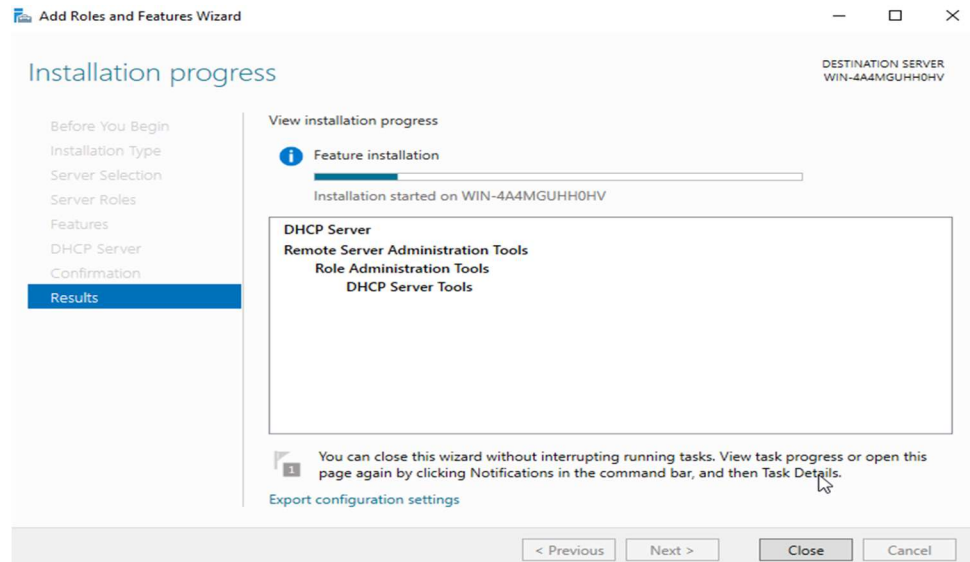
DHCP servers operate independently, so you must install the service and configure scopes on every computer that will function as a DHCP server.

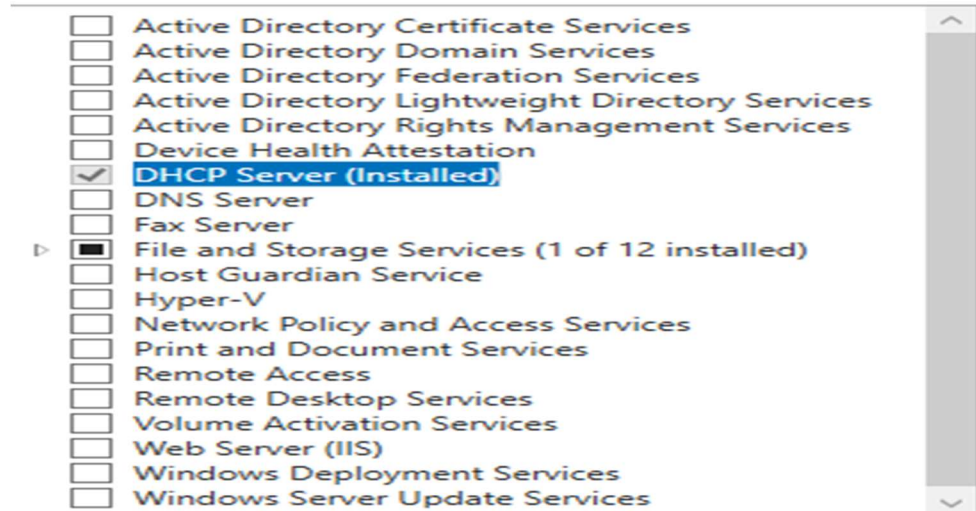
The DHCP Server service is packaged as a role in Windows Server 2022, which you can

install by using the Add Roles and Features Wizard, accessible from the Server Manager console.

To install the DHCP Server service on a Windows Server 2022 computer with Server Manager, use the following procedure.

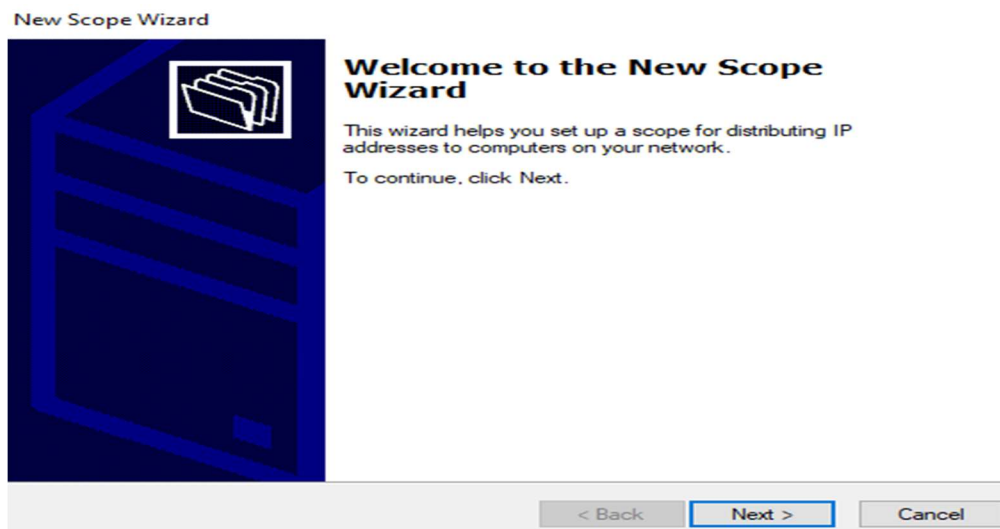
- In the Server Manager window, click Manage and then click Add Roles and Features. The Add Roles and Features Wizard appears, displaying the Before you begin page.
- Click Next. The Select Installation Type page appears.
- Leave the Role-based or feature-based installation radio button selected and click Next. The Select Destination Server page appears.
- Select the server on which you want to install the roles and/or features and click Next. The Select Server Roles page appears.
- Select the DHCP Server check box. An Add features that are required for DHCP Server. Dialog box appears.
- Click Add Features, and then click Next. The Select features page appears.
- Click Next. The DHCP Server page appears.
- Click Next. The Confirm installation selections page appears.
- Click Install. The Installation progress page appears as the wizard installs the role.
- Click Close. The wizard closes





3.2 Creating a DHCP Scope , Lease Durations and DNS server:

- In the Server Manager window, click Tools and then click DHCP. The DHCP console appears.
- Expand the server and IPv4 nodes.
- Right-click the IPv4 node and, from the context menu, select New Scope. The New Scope Wizard appears, displaying the Welcome page. Click Next. The Scope Name page appears.



- Type a name for the scope into the Name text box and click Next. The IP Address range page appears. The IP Address Range page in the New Scope Wizard appears displaying the welcome page.
- In the Start IP address text box, type the first in the range of addresses you want to assign. In the End IP address text box, type the last address in the range.
- In the Subnet Mask text box, type the mask value for the subnet on which the scope will operate and click Next. The Add Exclusions and Delay page appears.

- In the Start IP address and End IP address text boxes, specify a range of addresses you want to exclude from the scope. You can also specify a delay interval between the server's receipt of DHCPDISCOVER messages and its transmission of DHCPOFFER messages. Then click Next. The Lease Duration page appears.
- Specify the length of the leases for the addresses in the scope and click Next. The Configure DHCP Options page appears. Select Yes, I want to configure these options now and click Next. The Router (Default Gateway) page appears.

New Scope Wizard

Lease Duration

The lease duration specifies how long a client can use an IP address from this scope.

Lease durations should typically be equal to the average time the computer is connected to the same physical network. For mobile networks that consist mainly of portable computers or dial-up clients, shorter lease durations can be useful. Likewise, for a stable network that consists mainly of desktop computers at fixed locations, longer lease durations are more appropriate.

Set the duration for scope leases when distributed by this server.

Limited to:

Days: Hours: Minutes:

< Back Next > Cancel

- In the IP address text box, specify the address of a router on the subnet served by the scope and click Add. Then click Next. The Domain Name and DNS Servers page appears.

New Scope Wizard

Domain Name and DNS Servers

The Domain Name System (DNS) maps and translates domain names used by clients on your network.

You can specify the parent domain you want the client computers on your network to use for DNS name resolution.

Parent domain:

To configure scope clients to use DNS servers on your network, enter the IP addresses for those servers.

Server name: IP address:

Resolve

192.168.1.0

Add Remove Up Down

< Back Next > Cancel

- In the Server name text box, type the name of a DNS server on the network and click Resolve, or type the address of a DNS server in the IP address text box and click Add. Then click Next. The WINS Servers page appears.
- Click Next. The Activate Scope page appears.

- Select Yes, I want to activate this scope now and click Next. The Completing the New Scope Wizard page appears.
- Click Finish. The wizard closes.

Active Directory Configuration

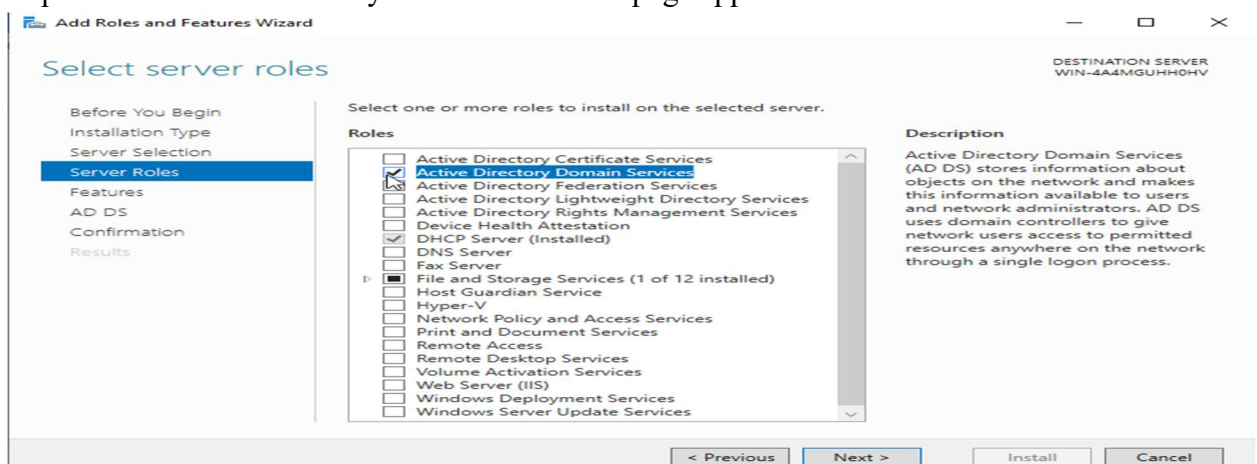
AD DS is a directory service that enables you to create organizational divisions called domains. A domain is a logical container of network components, hosted by at least one server designated as a domain controller. The domain controllers for each domain replicate their data among themselves for fault tolerance and load-balancing purposes.

To use a Windows Server 2022 computer as a domain controller, you must configure it to use static IP addresses, not addresses supplied by a Dynamic Host Configuration Protocol (DHCP) server. If you are creating a domain in an existing forest, or adding a domain controller to an existing domain, you also must configure the computer to use the DNS server that hosts the existing forest or domain, at least during the Active Directory installation.

4.1 Steps to setup Active Directory:

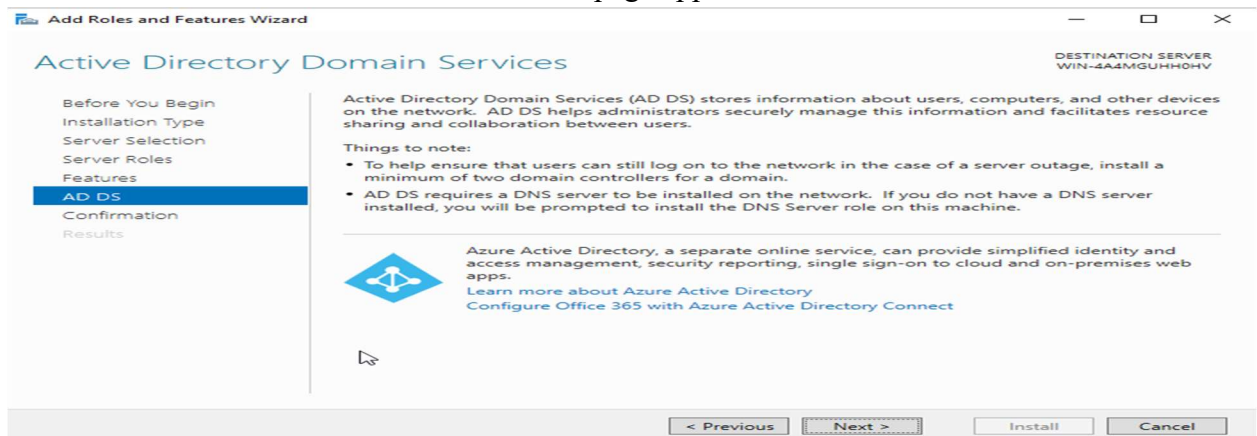
INSTALL THE ACTIVE DIRECTORY DOMAIN SERVICES ROLE

- From the Server Manager's Manage menu, select Add Roles and Features. The Add Roles and Features Wizard appears.
- Click Next. The Select Installation Type page appears.
- Leave the Role-based or feature-based installation radio button selected and click Next. The Select Destination Server page appears.
- Select the server that you want to promote to a domain controller and click Next.
- The Add Roles and Features Wizard displays the Select server roles page.
- Select the Active Directory Domain Service role. The Additional features that are required for Active Directory Domain Services? page appears.

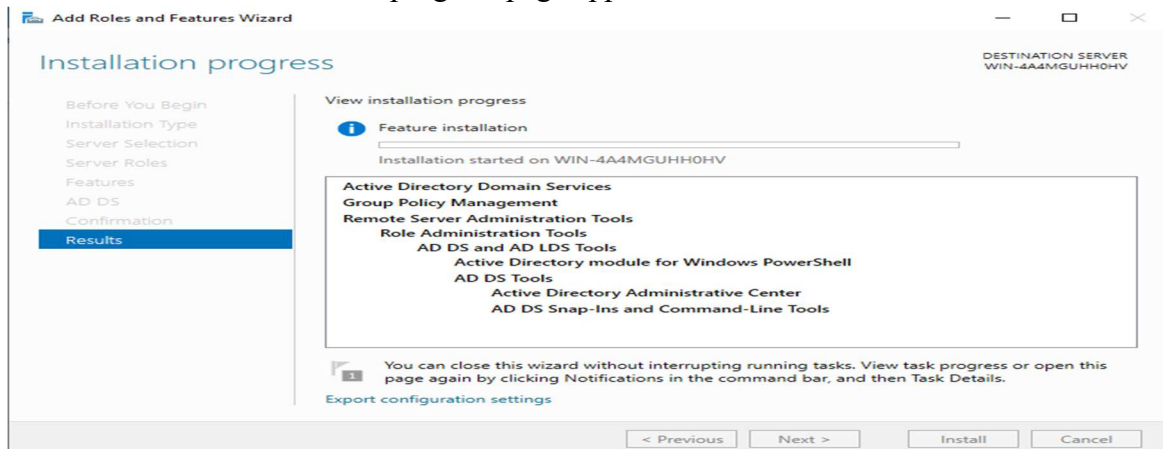


- Click Add Features to accept the dependencies, and then click Next. The Select features page appears.

- Click Next. The Active Directory Domain Services page appears, displaying information about the role.
- Click Next. A Confirm installation selections page appears.



- Select from the following optional functions, if desired:
 1. Restart the destination server automatically if desired causes the server to restart automatically when the installation is complete, if the selected roles and feature require it.
 2. Export configuration settings creates an XML script documenting the procedure.
 3. performed by the wizard, which you can use to install the same configuration on another server using Windows PowerShell.
 4. Specify an alternate source path species the location of an image file containing
 5. the software needed to install the selected roles and features.
- Click Install. The Installation progress page appears.



- After the role is installed, a Promote this server to a domain controller link appears.
- PAUSE. Leave the wizard open.

When beginning a new AD DS installation, you first need to create a new forest, which you do by creating the first domain in the forest, the forest root domain.

To create a new forest, use the following procedure:

- On the Installation progress page that appears at the end of the Active Directory Domain Services role installation procedure, click the Promote this server to a domain controller hyperlink. The Active Directory Domain Services Configuration Wizard appears, displaying the Deployment Configuration page.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard minimize, maximize, and close buttons. The main content area is titled 'Deployment Configuration'. On the left, there is a navigation pane with the following items: 'Deployment Configuration' (selected), 'Domain Controller Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area contains the following sections:

- 'Select the deployment operation' with three radio buttons: 'Add a domain controller to an existing domain', 'Add a new domain to an existing forest', and 'Add a new forest' (which is selected).
- 'Specify the domain information for this operation' with a text box labeled 'Root domain name:' containing the text 'helld'.
- A link at the bottom: 'More about deployment configurations'.

 At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

- Select the Add a new forest option and, in the Root domain name text box, type the name of the domain you want to create.
- Click Next. The Domain Controller Options page appears

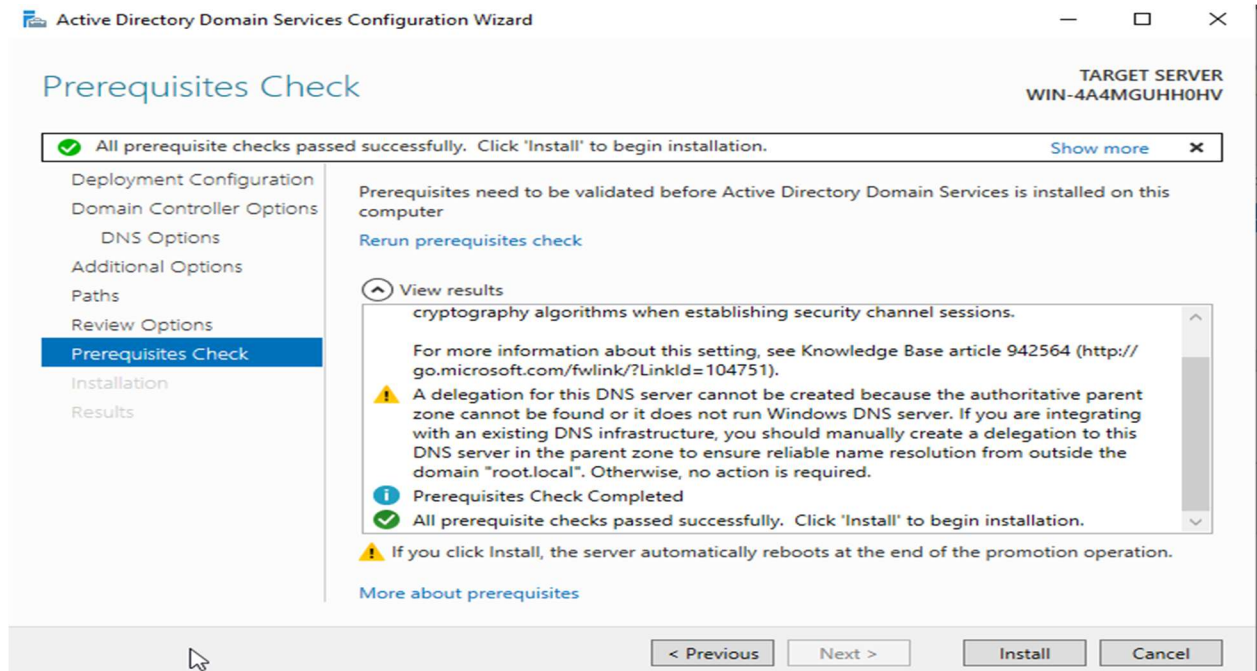
The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window, now on the 'Domain Controller Options' page. The title bar and window controls are the same. The navigation pane on the left now has 'Domain Controller Options' selected. The main content area contains the following sections:

- 'Select functional level of the new forest and root domain' with two dropdown menus: 'Forest functional level:' and 'Domain functional level:', both set to 'Windows Server 2016'.
- 'Specify domain controller capabilities' with three checkboxes: 'Domain Name System (DNS) server' (checked), 'Global Catalog (GC)' (checked), and 'Read only domain controller (RODC)' (unchecked).
- 'Type the Directory Services Restore Mode (DSRM) password' with two password fields: 'Password:' and 'Confirm password:', both with red asterisks indicating they are required.
- A link at the bottom: 'More about domain controller options'.

 At the bottom of the window, there are four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

- Make changes on the Domain Controller Options page, as needed:
 - If you plan to add domain controllers running earlier versions of Windows Server to this forest, select the earliest Windows version you plan to install from the Forest functional level drop-down list.
 - If you plan to add domain controllers running earlier versions of Windows Server to this domain, select the earliest Windows version you plan to install from the Domain functional level drop-down list.

3. If you do not already have a DNS server on your network, leave the Domain Name System (DNS) server check box selected. If you have a DNS server on the network and the domain controller is configured to use that server for DNS services, clear the check box.
- In the Password and Confirm password text boxes, type the password you want to use for Directory Services Restore Mode (DSRM) and click Next. The DNS options page appears, with a warning that a delegation for the DNS server cannot be created, because the DNS Server service is not installed yet.
 - Click Next. The Additional Options page appears displaying the NetBIOS equivalent of the domain name you specified.
 - Modify the name, if desired, and click Next. The Paths page appears.
 - Modify the default locations for the AD DS files, if desired, and click Next. The Review Options page appears.
 - Click Next. The Prerequisites Check page appears



- The wizard performs a number of environment tests to determine whether the system can function as a domain controller. The results can appear as cautions, which enable the procedure to continue, or warnings, which require you to perform certain actions before the server can be promoted.
- After the system passes all the prerequisite checks, click Install. The wizard creates the new forest and configures the server to function as a domain controller.
- RESTART the computer.

Adding a Domain Controller to an Existing Domain:

Steps to add a domain controller to existing domain. Every Active Directory domain should have a minimum of two domain controllers. To add a domain controller to an existing Windows Server 2022 domain, use the following procedure.

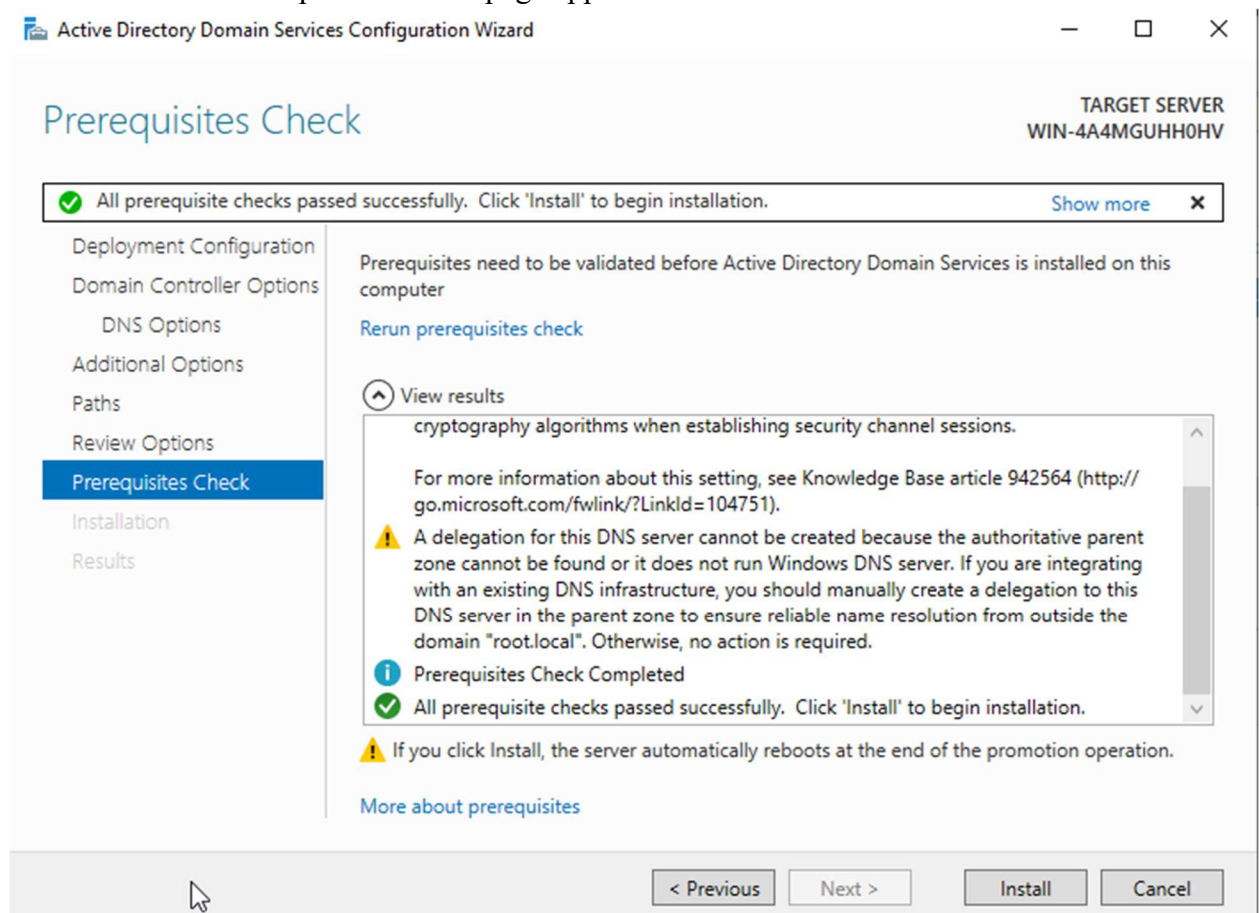
- On the Installation progress page that appears at the end of the Active Directory Domain Services role installation procedure, click the Promote this server to a domain controller hyperlink. The Active Directory Domain Services Configuration Wizard appears, displaying the Deployment configuration page. Select the Add a domain controller to an existing domain option and then click Select.

The screenshot shows the 'Active Directory Domain Services Configuration Wizard' window. The title bar includes the Windows logo and the text 'Active Directory Domain Services Configuration Wizard'. The window has standard minimize, maximize, and close buttons. The main content area is titled 'Deployment Configuration'. On the left, there is a navigation pane with the following items: 'Deployment Configuration' (highlighted), 'Domain Controller Options', 'Additional Options', 'Paths', 'Review Options', 'Prerequisites Check', 'Installation', and 'Results'. The main area contains the following sections: 1. 'Select the deployment operation' with three radio buttons: 'Add a domain controller to an existing domain' (selected), 'Add a new domain to an existing forest', and 'Add a new forest'. 2. 'Specify the domain information for this operation' with a 'Domain:' label and a text box containing a red asterisk, followed by a 'Select...' button. 3. 'Supply the credentials to perform this operation' with the text '<No credentials provided>' and a 'Change...' button. At the bottom of the main area is a link: 'More about deployment configurations'. The bottom of the window features a navigation bar with four buttons: '< Previous', 'Next >', 'Install', and 'Cancel'.

- If you are not logged on to an existing domain in the forest, a Credentials for deployment operation dialog box appears, in which you must supply administrative credentials for the domain to proceed. After you are authenticated, a Select a domain from the forest dialog box appears.
- Select the domain to which you want to add a domain controller and click OK. The selected domain name appears in the Domain field. Click Next. The Domain Controller Options page appears
- Select or clear the check boxes on the Domain Controller Options page as needed:
 - 1.If you want to install the DNS Server service on the computer, leave the Domain Name System (DNS) server check box selected. Otherwise, the DNS server the computer is configured to use will host the domain.
 - 2.Leave the Global Catalog (GC) check box selected if you want the computer to function as a global catalog server. This is essential if you are deploying the new domain controller at a site that does not already have a GC server.
 - 3.Select the Read only domain controller (RODC) check box to create a domain controller that administrators cannot use to modify AD DS

objects. In the Site Name drop-down list, select the site where the domain controller will be located.

- In the Password and Confirm Password text boxes, type the password you want to use for Directory Services Restore Mode (DSRM) and click Next. The Additional Options page appears
- To use the Install from Media (IFM) option, select the Install from media check box.
- In the Replicate from drop-down list, select the existing domain controller that the server should use as a data source. Then click Next. The Paths page appears.
- Modify the default locations for the AD DS files, if desired, and click Next. The Review Options page appears.
- Click Next. The Prerequisites Check page appears.



- After the system passes all the prerequisite checks, click Install. The wizard configures the server to function as a domain controller. Finally, RESTART the computer

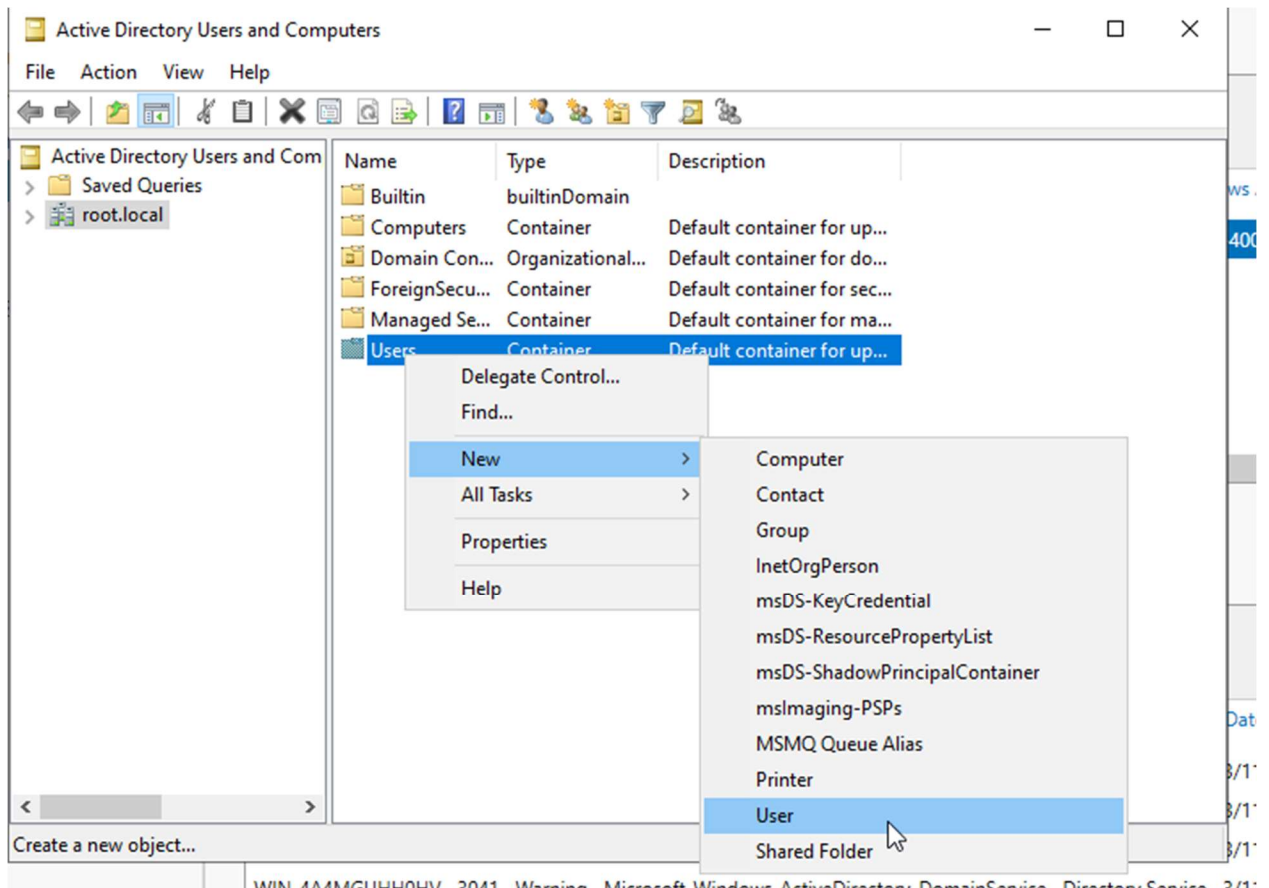
4.2 Creating a New Domain:

- After promoting the server to a domain controller, the new domain is automatically created.
- Ensure that the server has been assigned the appropriate IP configuration and DNS settings.

- Verify the new domain by opening Active Directory Users and Computers from Administrative Tools.
- Expand the "Domain" node to view the newly created domain.

4.3 Adding User Accounts:

- To add a new user account, open Active Directory Users and Computers.
- Right-click on the "Users" container or any specific OU where you want to add the user and select "New" > "User."



- Enter the user's First name, Last name, User logon name, and other required details.

The screenshot shows the 'New Object - User' dialog box with the title bar 'New Object - User' and a close button. Below the title bar is a header section with a user icon and the text 'Create in: root.local/Users'. The main area contains several input fields: 'First name:' with a text box, 'Initials:' with a text box, 'Last name:' with a text box, 'Full name:' with a text box, 'User logon name:' with a text box and a dropdown menu showing '@root.local', and 'User logon name (pre-Windows 2000):' with a text box containing 'ROOT\' and an empty text box. At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

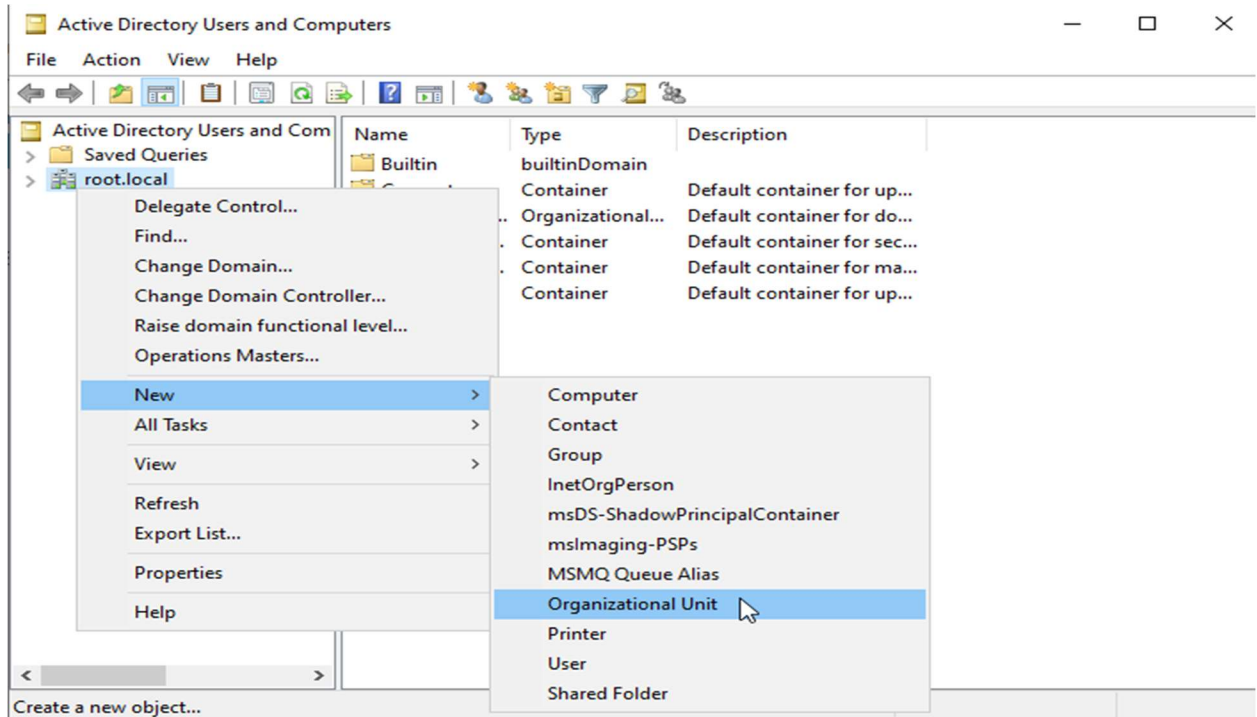
- Set a password for the user and choose appropriate password settings.

The screenshot shows the 'New Object - User' dialog box with the title bar 'New Object - User' and a close button. Below the title bar is a header section with a user icon and the text 'Create in: root.local/Users'. The main area contains two text boxes for 'Password:' and 'Confirm password:', followed by four checkboxes: 'User must change password at next logon' (checked), 'User cannot change password', 'Password never expires', and 'Account is disabled'. At the bottom are three buttons: '< Back', 'Next >', and 'Cancel'.

- Click "Next" and then "Finish" to create the user account.

4.4 Organizing Users into OUs:

- Organizational Units (OUs) help in organizing and managing users efficiently.
- To create a new OU, right-click on the domain or an existing OU, select "New" > "Organizational Unit."



- Enter the name of the OU and click "OK" to create it.
- Drag and drop user accounts organization from the "Users" container or other OUs into the newly created OU for better.

Connecting Computers to the Domain

5.1 Steps to join client computers to the domain created in Active Directory

1. Open System Properties:

- On the client computer, right-click on the "This PC" icon (Windows 10/11) or "My Computer" (Windows 7/8/Server editions) in File Explorer.
- Select "Properties" from the context menu to open the System Properties window.

2. Joining the Domain:

- In the System Properties window, click on the "Change settings" link next to the computer name.
- Click on the "Change" button in the System Properties dialog box under the "Computer Name" tab.
- In the Computer Name/Domain Changes dialog box, select the "Domain" option.
- Enter the name of the domain you want to join and click "OK."

- You will be prompted to provide domain administrator credentials. Enter the username and password of a user account with sufficient privileges to join computers to the domain.
- Click "OK" to join the domain.

5.2 Verifying Domain Membership:

- After providing domain credentials, the computer will attempt to join the domain. Once completed, you will receive a message indicating whether the operation was successful.
- If successful, you will be prompted to restart the computer to apply the changes. Click "OK" to restart the computer.
- After the computer restarts, log in using a domain user account to verify that the computer has successfully joined the domain.

5.3 To confirm domain membership:

- Right-click on "This PC" (Windows 10/11) or "My Computer" (Windows 7/8/Server editions).
- Select "Properties."
- In the System Properties window, under the "Computer name, domain, and workgroup settings," you should see the name of the domain listed.

5.4 Additional Verification:

- Open Active Directory Users and Computers on the domain controller.
- Navigate to the "Computers" container or the specific OU where the computer should be located.
- Verify that the computer object representing the joined computer appears in the list.

5.6. Repeat for Additional Computers:

- Repeat the above steps for each client computer you want to join to the domain.

User Profile and Home Folder Configuration:

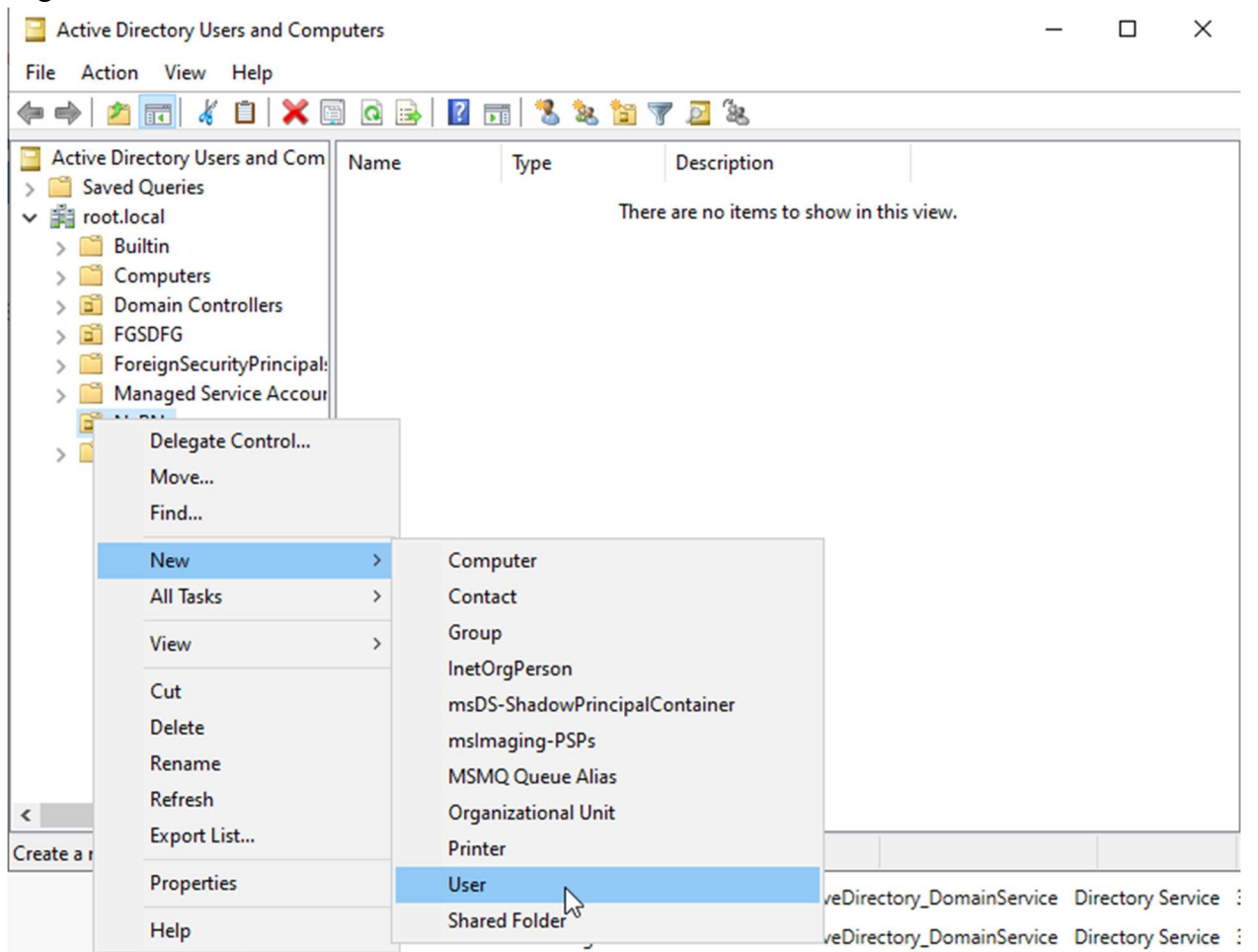
Configuring user profiles and home folders within Active Directory involves creating user accounts, assigning permissions, and setting up home folders for users to store their files securely. Below are the detailed steps to accomplish these tasks:

6.1 Steps to configure user profiles and home folders within Active Directory:

1. Creating User Accounts:

- Open Active Directory Users and Computers on the domain controller.
- Navigate to the container or Organizational Unit (OU) where you want to create the user account.

- Right-click on the container or OU and select "New" > "User."



- Enter the user's details such as First name, Last name, User logon name, and other necessary information.

The screenshot shows the 'New Object - User' dialog box. At the top, it says 'Create in: root.local/N ,BN,'. Below this, there are several input fields: 'First name:' with a text box containing 'I', 'Initials:' with an empty text box, 'Last name:' with an empty text box, and 'Full name:' with an empty text box. Below these is the 'User logon name:' section, which includes a text box and a dropdown menu currently showing '@root.local'. Underneath is the 'User logon name (pre-Windows 2000):' section, with a text box containing 'ROOT\' and an empty text box. At the bottom, there are three buttons: '< Back', 'Next >', and 'Cancel'.

- Set a password for the user account and configure any additional settings as needed.
- Click "Next" and then "Finish" to create the user account.

2. Assigning Permissions:

- Determine the appropriate level of access for users to their home folders.
- Right-click on the folder where you want to create home folders (e.g., a shared folder on a file server) and select "Properties."
- Go to the "Security" tab and click "Edit" to modify permissions.
- Click "Add" to add the user or group for which you want to assign permissions.
- Enter the name of the user or group and click "Check Names" to verify.
- Select the appropriate permissions for the user or group (e.g., Full Control, Modify, Read & Execute, etc.).
- Click "OK" to apply the changes and close the Permissions dialog box.

6.2 Setting Up Home Folders:

- Create a folder for each user's home folder within the shared folder or location you designated earlier.
- Right-click on the user folder you created and select "Properties."
- Go to the "Sharing" tab and click "Advanced Sharing."
- Check the box for "Share this folder" and click "Permissions" to set permissions for the home folder.
- Assign appropriate permissions for the user (e.g., Full Control, Change, Read) and click "OK" to apply the changes.
- Go to the "Security" tab and click "Edit" to modify permissions.
- Add the user or group that needs access to the home folder and assign appropriate permissions.
- Click "OK" to apply the changes and close the Properties dialog box.

6.3 Configuring User Profiles:

- Open Active Directory Users and Computers on the domain controller.
- Right-click on the user account you created earlier and select "Properties."
- Go to the "Profile" tab.
- In the "Profile" tab, you can specify the user's profile path. Enter the UNC path of the user's home folder (e.g., \\fileserver\username) as the profile path.
- Optionally, you can configure other profile settings such as profile path, logon script, and home folder settings.

- Click "Apply" and then "OK" to save the changes.

Basic Group Policy Implementation:

Implementing Group Policy Objects (GPOs) is a crucial aspect of network management in Windows Server environments. GPOs allow administrators to enforce standardized configurations across the network environment, including security settings, software installations, and other configuration requirements. Below are the steps to create and apply GPOs, along with examples of implementing basic group policies:

7.1 Steps to configure Group Policy:

1. Creation of Group Policy Objects (GPOs):

- Open the Group Policy Management Console (GPMC) on the domain controller.
- Navigate to the appropriate organizational unit (OU) or domain where you want to apply the GPO.
- Right-click on the OU or domain and select "Create a GPO in this domain, and Link it here."
- Enter a descriptive name for the GPO and click "OK" to create it.

2. Application of GPOs:

- Once the GPO is created, you can configure settings by right-clicking on the GPO and selecting "Edit."
- In the Group Policy Management Editor, navigate to the desired configuration settings under User Configuration or Computer Configuration.
- Configure the settings as needed using the available policies and preferences.
- After configuring the settings, close the Group Policy Management Editor.

File Sharing and storage Quota Setup:

Setting up file sharing on Windows Server is essential for facilitating seamless collaboration among users. Below are the steps to set up file sharing, create shared folders, configure permissions, and implement storage quotas:

8.1 Steps to Setup File Sharing

1. Setting Up File Sharing:

- Ensure that the File Services role is installed on the Windows Server. You can install this role using Server Manager.

- Open Server Manager and navigate to the "Shares" section under "File and Storage Services."
- Right-click on "Shares" and select "New Share" to create a new shared folder.

2. Creating Shared Folders:

- In the "New Share Wizard," select the type of shared folder you want to create (e.g., SMB Share or NFS Share) and click "Next."
- Choose the folder or volume that you want to share and click "Next."
- Enter a name and description for the shared folder and click "Next."
- Configure the share permissions (e.g., Read, Read/Write, etc.) and click "Next."
- Review the configuration summary and click "Create" to create the shared folder.

3. Configuring Permissions:

- After creating the shared folder, right-click on it and select "Properties."
- Go to the "Security" tab and click "Edit" to modify permissions.
- Add the appropriate users or groups and assign the necessary permissions (e.g., Full Control, Modify, Read, etc.).
- Click "OK" to apply the permissions.

8.2 Steps to Setup Storage Quota

- To implement storage quotas, open File Server Resource Manager (FSRM) from Server Manager.
- In FSRM, expand "Quota Management" and click on "Quotas."
- Right-click on "Quotas" and select "Create Quota."
- Select the volume where you want to apply the quota and click "OK."
- Specify the quota properties, such as the quota type (Hard quota or soft quota), the limit size, and the notification settings.
- Set the properties for the quota template, such as the name and description, and click "Create."
- Once the quota is created, it will be applied to the specified volume, and users will be restricted based on the configured quota settings.

5. Monitoring and Managing File Shares:

- Regularly monitor disk space usage and user access to shared folders using File Server Resource Manager (FSRM) reports and monitoring tools.
- Periodically review and adjust share permissions and storage quotas based on changing requirements and usage patterns.